Release Notes

22 October 2026

I. Supported cameras

Table I
Camera models

Camera	`camera.model`	Mono	Stereo
360° equirectangular (two lenses)	`EQUIRECTANGULAR`	✓	–
Pinhole, radial-tangential	`PINHOLE_RADTAN`	✓	✓
Fisheye, equidistant	`FISHEYE_EQUIDISTANT`	✓	✓

II. Features

- Online camera-IMU time offset estimation
- Robust tracking under auto-exposure changes and night lights
- 360°: online time offset estimation per lens
- 360°: online stitching error estimation

III. Experimental setup

Table II
Test environment

CPU	Intel Core i9-13900KF, **2 cores per run**
OS	Ubuntu 24.04
Datasets	PanoAir (17), EuRoC MAV (11), TUM-VI indoor (20)
Metric	ATE [m], SE(3) alignment
Baselines	OpenVINS, VINS-Mono (mono, no loop closure)

IV. Benchmark

Table III
Mean ATE [m] (completed / run), monocular

Dataset	Ours	OpenVINS	VINS-Mono
PanoAir, 17 flights	**1.14 (17/17)**	3.56 (8/17) *	4.17 (17/17) *
EuRoC MAV, 11	**0.122 (11/11)**	0.136 (10/11)	0.193 (11/11)
TUM-VI indoor, 20	**0.736 (20/20)**	1.064 (13/20)	1.062 (20/20)

Table IV
Run time on 2 cores, monocular (mean of six EuRoC MAV and TUM-VI sequences)

	Ours	OpenVINS	VINS-Mono
FPS	76.3	**94.3**	75.2
CPU usage [cores of 2]	1.31	**0.99**	1.59
Peak memory [MB]	**38.0**	68.6	116.0

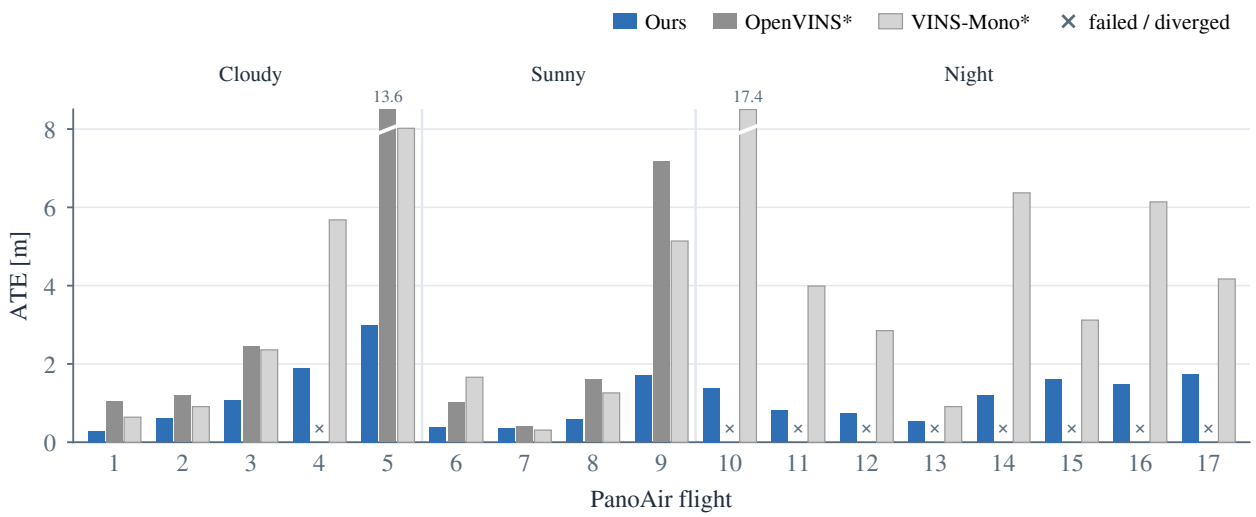


Fig. 1. ATE per PanoAir flight (360° camera on a drone, 16 km). ×: lost or diverged.

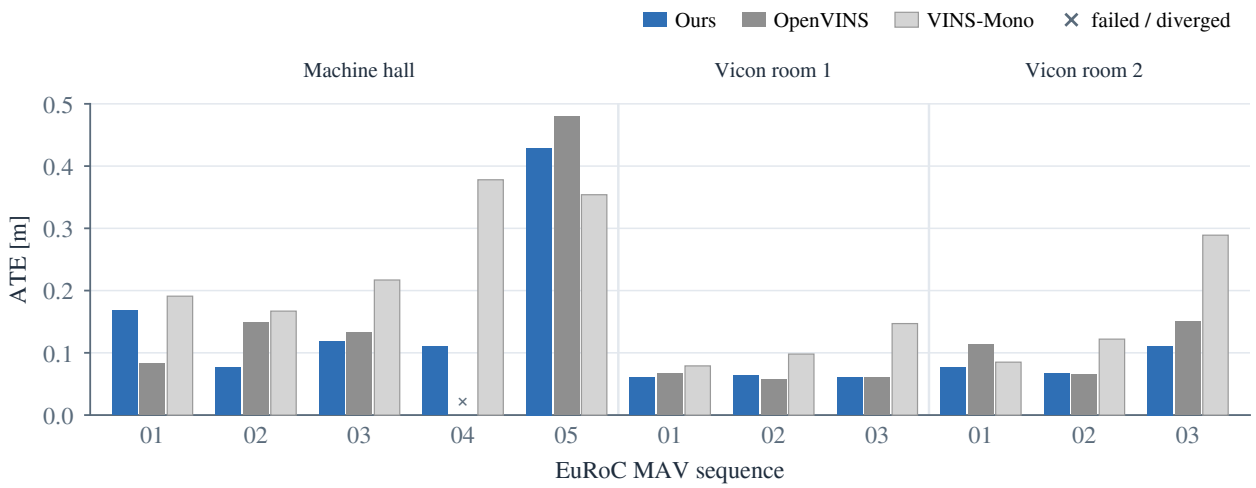


Fig. 2. ATE per EuRoC MAV sequence. ×: diverged.

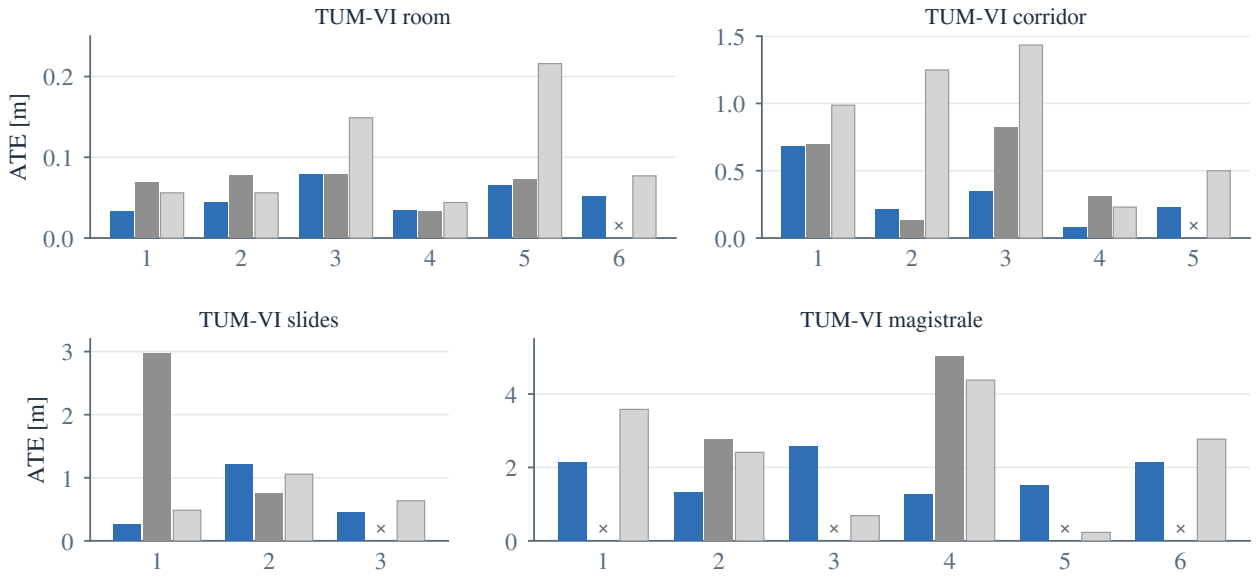


Fig. 3. ATE per TUM-VI indoor sequence (colors as in Fig. 2). ×: not initialized or diverged.

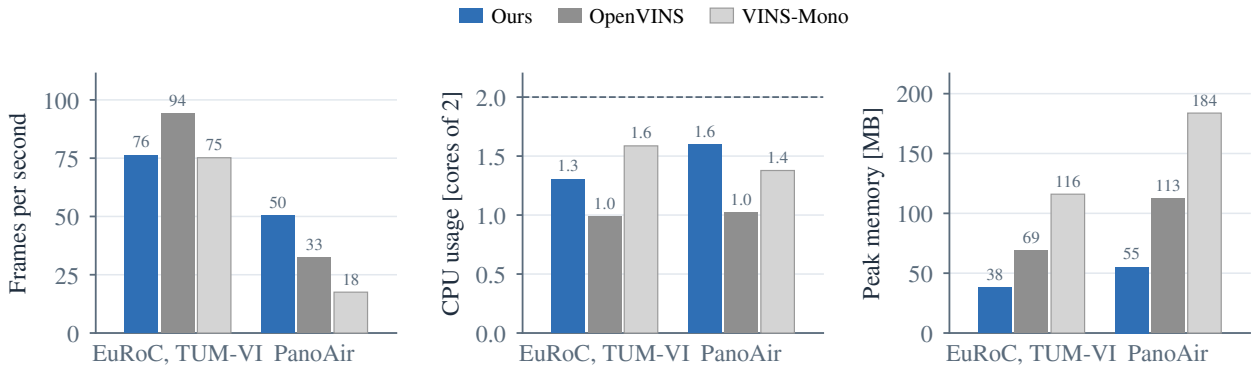


Fig. 4. Frames per second, average CPU usage (out of the 2 cores) and peak memory; PanoAir: three flights.

V. Stereo (extra)

Table V
Ours monocular vs. stereo on 2 cores

	Mean ATE [m]				
	EuRoC MAV	TUM-VI indoor	FPS	CPU [cores of 2]	Memory [MB]
Mono	0.122	0.736	76.3	1.31	38.0
Stereo	0.084	0.771	48.8	1.40	44.1